

Vortex-generator-induced intensification of fine mineral collection in pipe flow

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ABSTRACT

Flotation of fine minerals is becoming increasingly challenging. Mineral particles are believed to collide and attach efficiently to air bubbles in highly turbulent circular pipes. In this study, vortex generators (VGs) are adopted to strengthen the turbulence and improve the collection efficiency of fine minerals in pipe flow flotation. Computational fluid dynamics simulations, bubble size measurements, and flotation tests are conducted in a modified pipe unit with VGs and an initial pipe unit without VGs. The simulation results show that the VGs induce streamwise vortices and increase the turbulent dissipation rate in the pipe unit. Bubble size distribution measurements show that the VGs decrease the bubble size slightly at high velocities. Flotation tests prove that vigorous turbulence induced by VGs is beneficial to the recovery of fine particles. In particular, the flotation rate constant of $-20\ \mu\text{m}$ particles at a liquid velocity of $0.7\ \text{m/s}$ increases by more than 40% in the presence of VG-s5. Nevertheless, the excessive turbulence induced by VG-s5 caused a 20% decrease in the flotation rate of $+45\text{--}74\ \mu\text{m}$ particles when the liquid velocity increases to $2.0\ \text{m/s}$. This indicates that a relationship exists between turbulence and the particle size recovered. For a specified particle size, turbulence can be adjusted by varying the vortex generator spacing and changing the flow velocity to achieve the optimum flotation performance.

1. Introduction

The flotation beneficiation of fine minerals is a longstanding problem in mineral processing (Miettinen et al., 2010). The low mass of fine particles results in a low collision rate and efficiency with bubbles of a specified size and velocity (Trahar and Warren, 1976; Sivamohan, 1990; Dai et al., 2000). Existing industrial flotation plants fail to “recover” fine mineral particles below $20\ \mu\text{m}$, resulting in substantial losses of valuable raw material streams (Eckert et al., 2019). Several flotation technologies have been developed to increase bubble-particle collection efficiency, including selective flocculation (Panda et al., 2015; Song et al., 2001), hydrophobic aggregation (Rao, 1997), and microbubble flotation (Alam et al., 2017; Santana et al., 2012).

The flotation process can be modeled as a first-order rate process, which is expressed by

$$\frac{dN_p}{dt} = kN_p = -Z_{pb} \times E_{coll} = -Z_{pb} \times E_c \times E_a \times E_s, \quad (1)$$

where N_p is the number of solid particles; k is the flotation rate constant;

Z_{pb} is the number of collisions; E_c , E_a , and E_s refer to the efficiencies of the collision, attachment, and stability, respectively. In addition to particle size (Rulyov, 2001) and bubble size (Yoon and Luttrell, 1989), the hydrodynamic environment is vital to the bubble-particle collection process (Brabcová et al., 2015; Koh and Schwarz, 2006; Phan et al., 2003). Theoretical and numerical simulations indicate that the probability of collision and detachment increases with the turbulent intensity (Firouzi et al., 2011; Koh and Schwarz, 2003; Nguyen and Schulze, 2003; Wang et al., 2017a).

Mineral particles are believed to collide with the air bubbles and attach to them efficiently in highly turbulent circular pipes; hence, pipe flow flotation have been investigated extensively (Jameson, 1988; Li et al., 2020; Zhang et al., 2013). Li et al. (2009) summarized its application in the design of flotation columns. You et al. (2017) derived turbulent energy dissipation and utilized a flotation model to calculate the flotation rate constants of methylated quartz particles in a Jameson cell downcomer. Liao et al. (2014) and Yan et al. (2012) investigated the effects of pipe length on the flotation behavior of coal and bauxite, respectively. They concluded that the volume integral value of turbulent

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kinetic energy in a pipe unit grows logarithmically with an increase in the pipe length, and the recovery of fine minerals can be improved by increasing the turbulent intensity in the pipe unit.

The turbulent intensity can be increased to enhance the external energy input (Safari et al., 2016; Tabosa et al., 2016). However, this energetically expensive method results in increased production costs. Another effective method is to optimize the inner structure of the pipe unit using vortex generators (VGs). Numerical simulations and laboratory experiments suggest that VGs can enhance mixing performance by generating streamwise vortices (Hsiao et al., 2014; Mohand Kaci et al., 2009). Although the application of VGs in improving mixing efficiency has been emphasized, literature regarding the effect of VGs on mineral flotation is limited. In our previous study (Wang et al., 2017b), it was initially proven that VGs improved the combustible recovery of fine coal particles. However, the effect of VGs on the flotation behavior of different-sized particles has not been systematically investigated, rendering it difficult to modulate the flow field to fulfill the optimum turbulence conditions for the flotation of small-sized minerals, which hinders the application of VGs in the flotation industry.

In this study, vortex generators were adopted to increase the turbulent dissipation and improve the collection efficiency of fine minerals in pipe units. First, the pivotal flow field information in the modified pipe unit with VGs and the initial pipe unit were obtained via Computational fluid dynamics simulation. The effects of the vortex generator spacing distance and liquid flow velocity on the turbulent flow fields in the pipe unit were investigated. Subsequently, the bubble size and flotation rates of talc particles were measured and compared in the presence and absence of VGs. Based on numerical simulations and laboratory experimental results, the effect of turbulence induced by VGs on the flotation kinetics of different-sized particles is discussed. An effective turbulence optimization method for fine mineral flotation is proposed herein, and the relationship between turbulence induced by VGs and the particle size being recovered is revealed.

2. Numerical simulation

Based on our previous study (Wang et al., 2017b), VGs affect the flow field by changing the vortical structure and turbulent intensity in the pipe unit. In the present study, a spiral arrangement of the VGs was selected to enlarge the influence area of the vortex. The geometries of a VG and pipe unit are shown in Fig. 1. The geometric parameters were as follows: $a = b = 5$ mm; $h = 4.3$ mm; $\alpha = 45^\circ$; $\beta = 60^\circ$; $l = 200$ mm; $d =$

10 mm; $s = 5, 10$, and 15 mm for different pipe structures VG-s5, VG-s10, and VG-s15, respectively.

2.1. Simulation methodology

As the solid mass concentration of the particles in this study was 1% (corresponding to a solid volume concentration of 0.37%) and the particle size was less than $74 \mu\text{m}$ (which allows the particles to follow the slurry well), the pulp and bubbles were regarded as the homogeneous liquid and gas phases, respectively. Hence, gas–liquid two-phase simulations were conducted to obtain the flow characteristics of the pipe flow units. The Euler–Euler approach was used to describe the motion of the two continuous phases. In the present simulations, the momentum exchange between the two phases due to drag and buoyancy on the bubbles was the only mechanism that coupled the motion of the two phases. The Schiller–Naumann drag model was employed to calculate the drag coefficient of the bubbles. The renormalization group k – ϵ model, which is typically accepted in simulations of pipe flows with VGs, was applied to calculate the turbulent dissipation rate (Lei et al., 2017).

Unstructured grids were generated using the ANSYS ICEM software. The volume-averaged turbulent dissipation rate ($\bar{\epsilon}$) was selected as the main criterion for the grid-independence test of a single VG array. The relationship between the volume-averaged turbulent dissipation rate ($\bar{\epsilon}$) and the number of cells in each test is presented in Table 1. The volume-averaged turbulence dissipation rate ($\bar{\epsilon}$) increased by 3.21% as the number of cells increased from 1,802,672 (M4) to 2,210,768 (M5). Considering both the calculation accuracy and cost, mesh M4 was selected for the pipe flow simulation.

Using the software package ANSYS Fluent, simulations were conducted with liquid flow velocities of 0.7, 1.0, 1.5, and 2.0 m/s. The gas flow rate was set at a constant value of 1.0 L/min. The results for the pipe with VG-s5 at a liquid velocity of 1.0 m/s are presented herein to illustrate the flow structure yielded by the VG.

2.2. Flow pattern

Fig. 2 exhibits the velocity distribution at the cross section behind a VG. A high-speed region with a flow velocity of 1.4 m/s was formed in the center, compared with a flow velocity of 0.2 m/s region immediately behind the VG. A counter-rotating vortex pair (CVP) caused by the pressure gradient can be observed. The CVP can significantly enhance the flow mixing, thereby resulting in the suspension of solid particles and the dispersion of air bubbles, which is beneficial to the bubble–particle collision in the flotation process.

2.3. Turbulent dissipation rate (ϵ)

The turbulent dissipation rate (ϵ) is the energy dissipation caused by the viscous shear of the fluid. It is governed by the viscosity and normal gradient of the fluctuation velocity of the fluid, expressed as follows:

$$\epsilon = \frac{1}{2} \nu \left(\frac{\partial u_i'}{\partial x_j} + \frac{\partial u_j'}{\partial x_i} \right)^2, \quad (2)$$

where ν is the kinetic viscosity. ϵ can be employed to calculate the bubble–particle collision rate and efficiency (Koh and Schwarz, 2003).

The distribution of ϵ on the cross section was extracted every 5 mm along the flow direction. As shown in Fig. 3, the turbulent dissipation

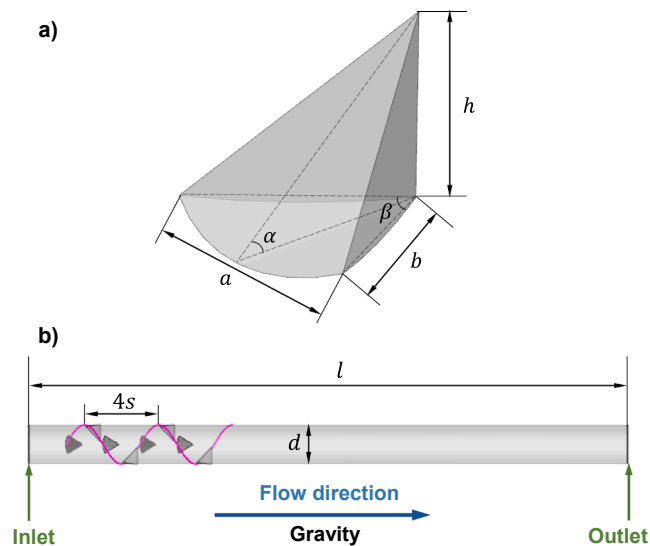


Fig. 1. (a) Geometry of VG and (b) spiral arrangement of VGs in pipe unit. Single-column.

Table 1

Volume-averaged turbulent dissipation rate ($\bar{\epsilon}$) for various numbers of cells.

Mesh	M1	M2	M3	M4	M5
Number of cells	227,136	712,153	1,106,734	1,802,672	2,210,768
Vol-avg tur-dis- rate (m^2/s^3)	10.57	14.86	17.81	20.53	21.19

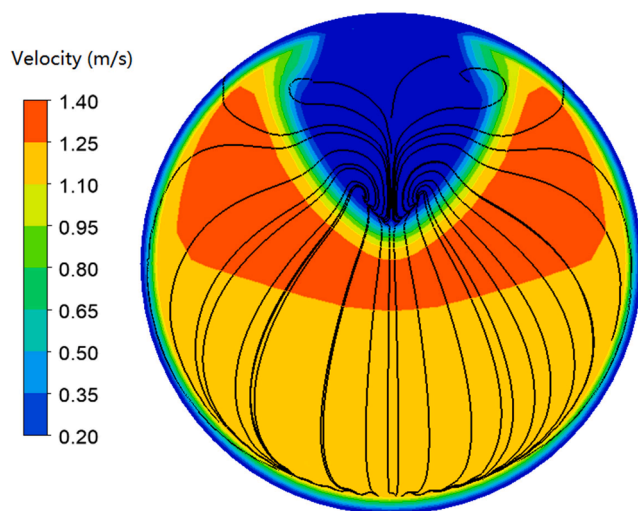


Fig. 2. Velocity distribution at cross section behind VG. Single-column.

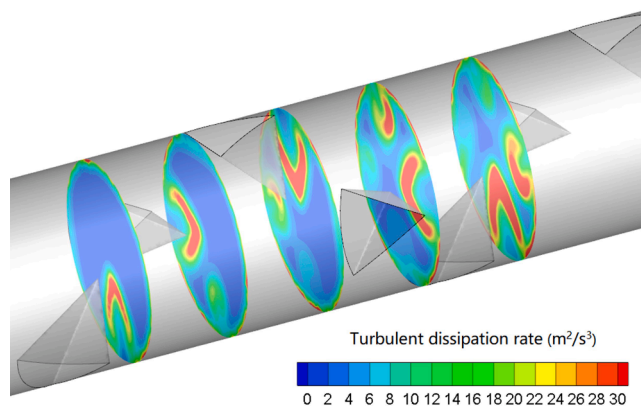


Fig. 3. Turbulent dissipation rate distribution on cross section along flow direction. Single-column.

rate enhanced continuously as the number of VGs increased. The streamwise vortices formed behind the spirally arranged VG interacted with each other and strengthened the turbulent intensity in the pipe unit. The $\bar{\epsilon}$ for all the conditions is shown in Fig. 4. The VGs significantly

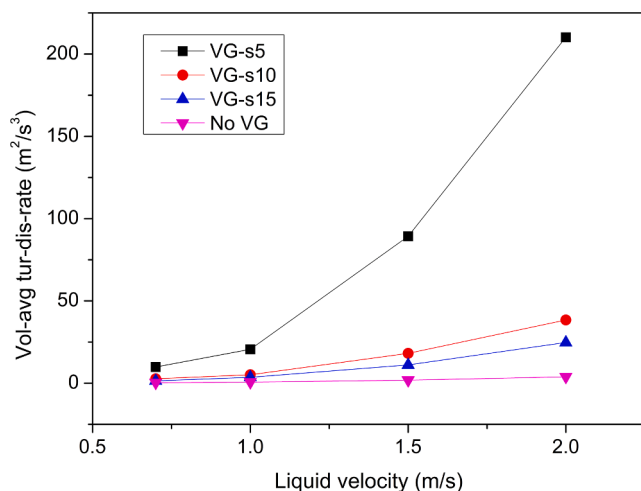


Fig. 4. Volume-averaged turbulent dissipation rate as a function of liquid velocity for different pipe structures. Single-column.

increased the turbulent dissipation rate in the pipe unit. The sequence of $\bar{\epsilon}$ was VG-s5 > VG-s10 > VG-s15 > No-VG. For all pipe structures, $\bar{\epsilon}$ increased with the liquid velocity. The smaller the spacing of the VG, the faster $\bar{\epsilon}$ increased with the liquid velocity. For the pipe with VG-s5, the $\bar{\epsilon}$ increased from 9.83 to 210.28 m^2/s^3 when the liquid velocity was increased from 0.7 to 2.0 m/s. This is because the presence of VGs in close proximity caused frequent flow separation and shearing in the flow at higher velocities.

3. Experiments

3.1. Materials

A natural hydrophobic mineral, talc, was selected to eliminate the effects of variable hydrophobicity and collector efficiency in the flotation experiments. A high-purity talc sample (99%), obtained from Guangxi Province, China, was ground in a laboratory stainless steel rod mill and passed through a 74 μm sieve. The $-74 \mu\text{m}$ samples were screened into -20 , $+20-45$, and $+45-74 \mu\text{m}$. Their X-ray diffraction results are shown in Fig. 5. The contact angle of these particles was determined to be 73.6° using Washburn's method on a Tensiometer K100 (KRUS, Germany). 4-Methyl-2-pentanol (MIBC) of analytical grade quality was used as the frother. Deionized water was used in all experiments.

3.2. Determination of bubble size

As shown in Fig. 6, a high-speed camera system was used to measure the bubble size distribution (BSD). The bubble observation chamber was set at an angle of 75° with respect to the sampling tube, and a 5-mm-thick observation chamber was used to prevent the overlapping and adhesion of bubbles. The port of the bubble sampler was installed 5 cm below the interface to avoid disturbance. Both the observation chamber and bubble sampler were filled with a deionized aqueous solution with a concentration of 50 ppm MIBC (the same concentration as in the flotation).

Bubbles were extracted from the separation column and transferred to a bubble observation chamber along the sampling tube. A CCD camera with 500 fps was used to capture the bubbles under certain lighting conditions. The bubble size was analyzed using the software package Image-Pro Plus (version 7.0; Media Cybernetics, USA). The capture range was $5.0 \text{ mm} \times 5.0 \text{ mm}$, and the resolution of each image was $1280 \text{ pixels} \times 1024 \text{ pixels}$, which corresponds to a spatial resolution of $4.89 \mu\text{m}$ per pixel. The maximum error of edge identification was ± 0.5 pixel. Hence, a maximum diameter error of ± 1 pixel, i.e., $\pm 4.89 \mu\text{m}$, was obtained. The Sauter mean diameter was calculated because it is

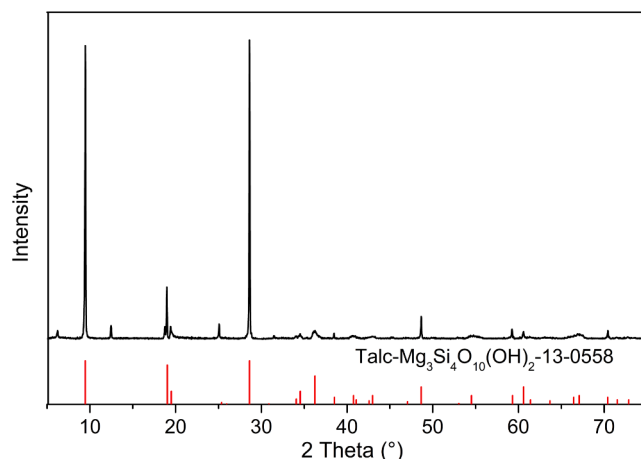


Fig. 5. X-ray diffraction spectrum of talc sample. Single-column.

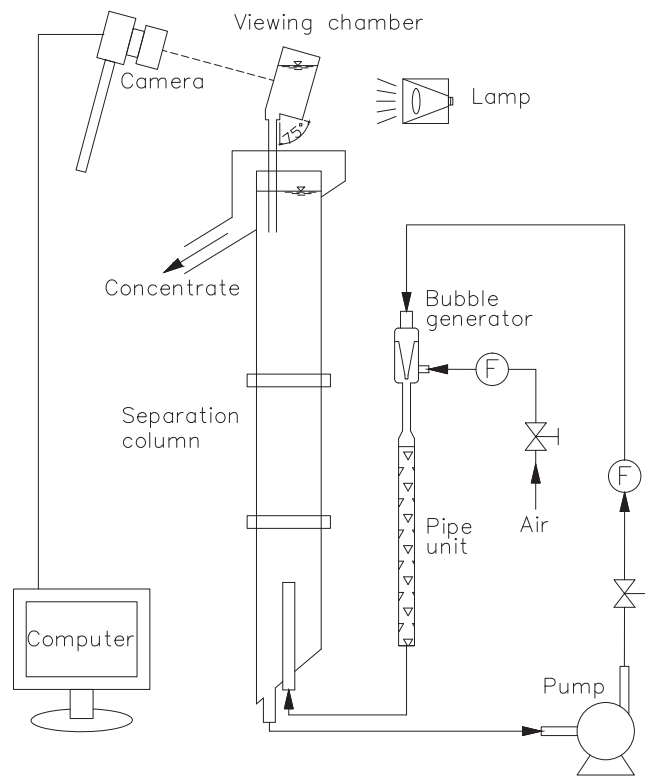


Fig. 6. Experimental setup of flotation tests and bubble size measurements. Single-column.

associated closely with the flotation rate. It is defined as

$$d_{32} = \frac{\sum_{i=1}^m d_i^3}{\sum_{i=1}^m d_i^2} \quad (3)$$

where d_i is the bubble diameter, and m is the number of bubbles measured. In this study, 300–500 bubbles were counted for each condition.

3.3. Flotation kinetics tests

Flotation kinetics tests were performed using a laboratory flotation column (Fig. 6). For each test, the conditioned suspension was poured into the flotation column and circulated using a pump. The liquid flow rate was controlled using a throttling valve. Air was absorbed by a Venturi-type bubble generator and then regulated by a needle valve. A turbine flow meter and magnetic flow meter were used to measure the liquid and air volumetric flow rates, respectively. Bubbles interacted with particles in the pipe unit and subsequently flowed into the separation column, forming a thin froth layer. All froth products were acquired from the top of the pulp as the concentrate. The operating

Table 2
Flotation column design and operating parameters.

Parameter	Value
Column internal diameter (mm)	60
Column height (mm)	550
Pipe internal diameter (mm)	10
Pipe length (mm)	200
Pipe internal structure	VG-s5, VG-s10, VG-s15, No VG
Pipe flow velocity (m/s)	0.7, 1.0, 1.5, 2.0
Solid concentration (mass%)	1
MIBC concentration (ppm)	50
Airflow rate (L/min)	1.0
Froth thickness (mm)	5

parameters of the flotation system and their values are listed in Table 2.

Flotation kinetic tests were performed on $-20 \mu\text{m}$, $+20-45 \mu\text{m}$, and $+45-74 \mu\text{m}$ talc particles. During all these processes, five concentrates were acquired at 30, 60, 120, 180, and 300 s and then placed in separate containers. All flotation products were filtered, dried, and weighed. The mass recovery was calculated and fitted to the following first-order rate equation:

$$R(t) = R_{\max}(1 - \exp(-kt)), \quad (4)$$

where $R_{\max}$ is the ultimate recovery, and k the flotation rate constant. It is noteworthy that entrainment was assumed to be negligible owing to the low solid concentration and water recovery. The froth depth was maintained below 5 mm; hence, the froth recovery was assumed to be 100% in this study. Reproducibility was assessed by performing three replicate tests under all the conditions.

4. Experimental results and discussion

4.1. BSD

Fig. 7 shows the cumulative BSD and Sauter mean bubble diameter (d_{32}) in the absence and presence of VG-s5 at a constant gas flow rate (1 L/min) and various pipe flow velocities, i.e., 0.7, 1.0, 1.5, and 2.0 m/s. As shown in Fig. 7, increasing the flow velocity resulted in a smaller bubble size both in the absence and presence of the VGs, which is consistent with published data (Yin et al., 2015). The percentage of small bubbles increased significantly when the flow velocity increased from 0.7 to 2.0 m/s; for example, the percentage of bubbles below $400 \mu\text{m}$ increased from 42.15% to 78.09%. The d_{32} decreased significantly from 508.59 to $426.87 \mu\text{m}$ in the absence of the VGs. For a specified Venturi-type bubble generator, the higher the flow velocity in the throat, the greater is the turbulent intensity in the expansion section, which directly facilitates bubble breakup (Yin et al., 2015; Zhao et al., 2017).

As for the effect of the VGs on the bubble size, the change in the d_{32} value at low speeds with and without the VGs was approximately $10 \mu\text{m}$. Considering the error in measuring individual bubbles, this difference can be regarded as an experimental error. However, the presence of VGs reduced the bubble d_{32} by $30 \mu\text{m}$ at high velocities. Based on the numerical simulation results shown in Fig. 4, the VG-s5 increased the $\bar{\epsilon}$ of the pipe unit to 89.30 and $210.28 \text{ m}^2/\text{s}^3$ at 1.5 and 2.0 m/s, respectively. This intense turbulent environment provided a secondary shear fragmentation of bubbles flowing through the pipe section.

It is noteworthy that this reduction is relatively small compared with the significant decrease caused by the increase in the flow velocity. This is likely because the turbulent intensity in the pipe unit was lower than that in the bubble generator, and the fluid characteristics of the latter

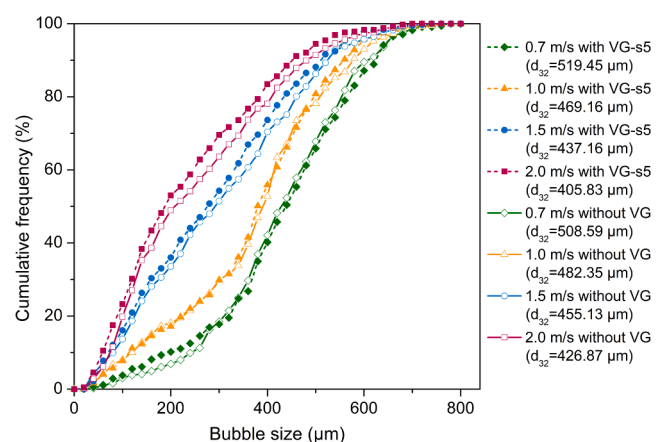


Fig. 7. BSD as a function of pipe flow velocity in presence of VGs (dashed lines) and absence of VGs (continuous lines). Single-column/1.5-column.

were key factors affecting the BSD.

4.2. Flotation test results

To understand the flotation behavior of talc with different particle sizes, flotation kinetics tests of $-20\ \mu\text{m}$, $+20-45\ \mu\text{m}$, and $+45-74\ \mu\text{m}$ particles were performed. The flotation rate constants for different pipe structures under different hydrodynamic conditions are shown in Fig. 8.

For the $-20\ \mu\text{m}$ particles, the flotation rate constant k increased with the flow velocity. In fact, a maximum increase of 230% was observed when the liquid velocity increased from 0.7 to 2.0 m/s. This significant increase suggests that energy input benefits the flotation of fine particles, as described in previous studies (Cheng, 2014; Taşdemir et al.,

2007; Safari et al., 2016). This is due to an increase in the bubble-particle collision probability. Moreover, the presence of the VGs increased the flotation rate. This improvement became more pronounced as the VG spacing decreased. For instance, compared with the pipe without VGs, the value of k for a pipe with VG-s5 increased by more than 40% at a liquid velocity of 0.7 m/s. The VGs in close proximity generated a stronger turbulent environment than the other structures, thereby facilitating the collision between the bubbles and fine particles, resulting in a higher flotation rate.

For the $+20-45\ \mu\text{m}$ particles, at low velocities of 0.7 and 1.0 m/s, the sequence of flotation rate was as follows: $k_{\text{VG-s5}} > k_{\text{VG-s10}} > k_{\text{VG-s15}} > k_{\text{No-VG}}$, which is consistent with the $-20\ \mu\text{m}$ particles. Increasing the liquid velocity improved the talc flotation kinetics. However, the intensification of VG-s5 on the flotation rate was no longer significant at high flow velocities. In fact, some of these particles detached from the bubbles because of the strong turbulence in the VGs-equipped pipe at 2.0 m/s, which resulted in a lower flotation rate ($1.48\ \text{min}^{-1}$) compared with that of the pipe without VGs ($1.66\ \text{min}^{-1}$). By contrast, the pipe structures with VG-s10 and VG-s15 exhibited better flotation performances. Based on the numerical simulation results, even at a high velocity of 2.0 m/s, the $\bar{\varepsilon}$ of the pipe structure with VG-s15 and VG-s10 remained less than $40\ \text{m}^2/\text{s}^3$, as compared with $210.28\ \text{m}^2/\text{s}^3$ for the structure with VG-s5.

For the $+45-74\ \mu\text{m}$ particles, the flotation rate correlated positively with the pipe flow velocity for the pipe structures. However, in the presence of VG-s5, a 5% lower flotation rate was obtained when the liquid velocity increased from 1.5 to 2.0 m/s. The presence of VG-5 was detrimental to the flotation at high pipe flow velocities. The comparative decrease of k at 2.0 m/s was approximately 20%. As shown by the numerical simulation results, the $\bar{\varepsilon}$ increased to $210.28\ \text{m}^2/\text{s}^3$ at 2.0 m/s liquid velocity. A high $\bar{\varepsilon}$ indicates that the particles on the bubble is affected by a strong centrifugal force that tends to move away from the bubble (Pyke, et al., 2003). The excessive turbulent environment caused the detachment of particles, resulting in a decrease in the flotation rate (Testa, et al., 2017; Safari, et al., 2017; Tabosa, et al., 2016).

The flotation kinetics of three class sizes under the same fluid environment (i.e., the same flow velocity and same structure of the pipe unit in this study) were investigated. The sequence of the flotation rate was $k_{+45-74} > k_{+20-45} > k_{-20}$. This is due to an increase in the probability of bubble-particle collisions for larger particles, which is conducive to bubble-particle acquisition. For a specified hydrodynamic environment, the $+45-74\ \mu\text{m}$ particles were more likely to detach from the bubbles than the finer particles. The flotation results showed a relationship between the turbulent intensity and the particle size being recovered. In other words, different hydrodynamic environments (in term of the turbulent dissipation rate and vortex distribution) must be established to recover particles of different sizes.

The turbulent flow can be controlled by varying the spacing of the VGs and changing the flow velocity. The VGs in close proximity, i.e., the pipe with VG-5, was used to generate an extremely strong turbulent flow field, which significantly enhanced the recovery of fine grains ($-20\ \mu\text{m}$) by promoting collision. However, detachment occurred among the larger particles. Hence, a pipe with VG-s10 is the preferred choice for coarser particles. It is noteworthy that VG-s10 the preferred choice among the four types of pipe structures investigated in this study. To obtain the optimum pipe structure, more geometric models and arrangements of the VG should be investigated.

5. Conclusions

In this study, VGs were used to improve the flotation performance of fine minerals. The turbulence in the pipe unit induced by the VGs and their effects on the flotation kinetics of different-sized particles were numerically and experimentally investigated. The simulation results showed that the VGs induced streamwise vortices and increased the turbulent dissipation rate in the pipe unit. The flotation tests proved that

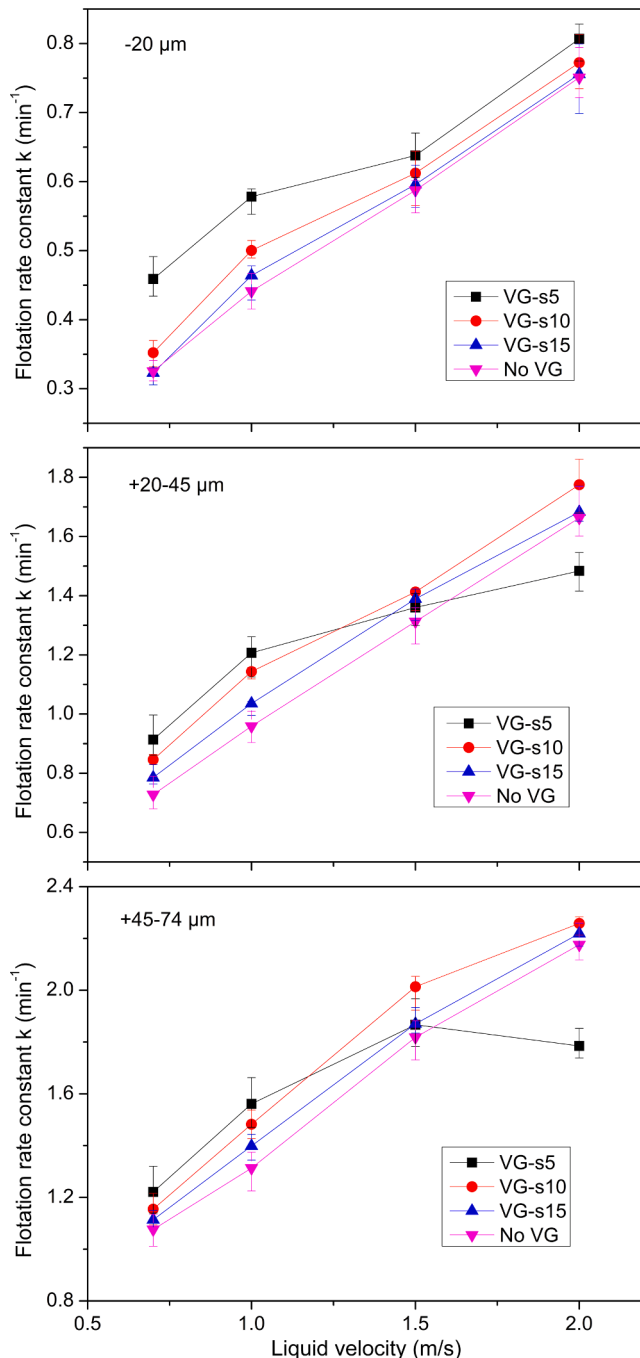


Fig. 8. Flotation rate constant of three particle sizes vs. liquid velocity for different pipe structures. Single-column.

the vigorous turbulence induced by the VGs facilitated the recovery of fine particles. This suggests that turbulence optimization using VGs is a potential solution to the problem of recovering fine-grained minerals.

Furthermore, the results indicated a relationship between turbulence and the particle size being recovered. For a specified particle size, the turbulence can be adjusted by varying the VG spacing and changing the flow velocity to achieve the optimum flotation performance. The method proposed for improving the recovery of fine minerals by controlling turbulence using VGs may provide valuable insights into the design and development of pipe flow flotation devices.

CRedit authorship contribution statement

Lijuan Sun: Methodology, Formal analysis, Investigation, Writing - original draft. **Yijun Cao:** Conceptualization, Supervision. **Lijun Wang:** Writing - review & editing. **Kaixin Zheng:** Validation, Formal analysis, Investigation. **Xiaokang Yan:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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